Team Project Report

Stack Overflowed

March 26, 2026

Introduction

Summary
Footprint is a web application designed to reduce carbon dioxide emissions by fostering awareness through a challenge-based, competitive platform. Users can complete daily, weekly, and monthly challenges across a range of environmental categories, each aimed at lowering their personal carbon footprint. Progress is quantified by tracking the amount of carbon saved, which is then converted into points; enabling users to compete against one another and harness the power of healthy competition to drive meaningful environmental impact.

Addressing the Problem
Carbon dioxide is the leading greenhouse gas that impacts our society today; it is the driving factor for climate change and global warming. Data from 2022 shows that carbon dioxide is responsible for 80% of total greenhouse gas emissions in that year [5], which shows how important it is for us to reduce our carbon footprint. Together, we can find a way to reduce our carbon footprint! This application is our attempt to do exactly that, by giving users different and engaging ways to reduce their footprints through a gamified challenge-based system. The audience for this application is primarily students and staff at the University of Exeter, with potential scope for future development in other scenarios too. For example, other universities, secondary schools and potentially workplaces. The application will also be able to support moderators, who will assign challenges and also review the submissions of users' completed challenges. This is further explained in the following sections.

Our focused goals:

- Log the micro changes of carbon data and simulate data for analytical purposes.
- Reduce carbon emissions and promote eco-friendly habits across users.
- Create a fun way to introduce sustainability with a gamified experience.

Our targeted stakeholders:

- University students across all years as well as faculty members, focusing on producing a fun and engaging product which will captivate their interest and prolonged engagement.
- The universities implementing the product want to find a way to engage their students in eco-friendly practices as well as reach targets on net-zero carbon emissions.

Success measures
Reflecting on our goals, we have chosen to measure the success of our product in accuracy of our data analysis from the recorded user activity, as well as the user engagement achieved from gamifying the experience. However, as this initial sprint focuses on producing a minimal working product this sprint will be solely evaluated by the runtime and availability of our prototype as well as minimising errors and bugs interacting with the system.

How will this be done and maintained?
This will be accomplished by users uploading pictures as a form of evidence for certain tasks they have completed. The moderator will verify the evidence and can approve or deny the submission's request. If the submission is approved, the user will get the points as per the pre-assigned points' value from a table that contains the points for that specific task. If the user's submission is denied, then the user will not get any points for that activity.

Prioritised Requirements

This section will be outlining the requirements of the prototype system, discussing both functional and non-functional requirements. Using the MoSCoW prioritisation method, we categorised features into must have, should have, could have and won't have. We were able to identify the Minimum Viable Product (MVP) for the first sprint. Requirements were categorized on the basis of their necessity for the system functionality and their contribution to meeting the objectives of the first sprint.

User Stories and Acceptance Criteria

Number	User Story	Acceptance Criteria
1	As a user, I want to sign up/sign in easily.	User can sign up with their username and password of choice, and users can sign in with their username and password.
2	As a user, I want to see the challenges I can do and the points I get for completing said challenge.	Table containing challenges users can complete and the points they can get for doing it.
3	As a user, I want to easily see the leaderboard and my ranking.	A table with a list of users and how many points they have in an order of highest number of points to lowest.
4	As a moderator, I want a specific and clear space where I can easily approve or deny user evidence.	A moderator only section where they can see user submission with the option of approving their submission and user gets points for the challenge or deny user submission and user gets no points.
5	As a user, I want the submission process to be clear and know how many points I will get from the challenge done.	A clear calculation showing user how the points are derived for the specific task they are doing.
6	As a moderator, I want to have a specific tab where I can easily create/end challenges and view the analytics of each challenge.	A tab specific for moderators only where they can make challenges for users and remove previous challenges.

Carbon Calculation model

The model design was shaped by balancing computational simplicity and user engagement with a gamified web application. The primary goal was to provide an approximate for the user rather than a fully personalised carbon footprint assessment, as this would require much more input which would make users perhaps less inclined to use the service, and also makes it easier to produce as there is less input to deal with.

Data figures, sources and relevance.
The values for carbon emissions were picked from publicly available carbon calculators and datasets about the environment. These sources provided data on emissions in different contexts hence, had to be normalised to a per-action estimate to ensure consistency across all challenges. An example of this is that the emissions from a reusable cup were divided by the average lifespan of uses of the cup thereby we came to a conclusion of 500 uses and the emissions from recycling were converted into daily values from distributing monthly averages. To maintain a uniform measurement scale, all the carbon savings were expressed as kilograms of CO_2 ($kgCO_2$). Doing this was essential for standardisation to be able to have leaderboard comparisons and overall scoring logic.

Assumptions and model limitations
The assumptions which the model relies on to be simple.

- Three meals per day when estimating vegetarian meal savings
- Average petrol vehicle emissions per kilometre
- Full daily recycling behaviour
- One repaired item per month
- Average food weight in a Too Good To Go purchase

We had to make assumptions as we did not not have enough data on users to make a specific lifecycle. Although there is a trade-off, while this does reduce individual user accuracy, it enhances usability and reduces excessive data collection, which would negatively affect user adoption and make the system more complex.

Categorisation of frequency of activities and actions.
The possible actions were categorised into 3 groups, daily, weekly or monthly. This was done to reflect real world behaviours and lifestyles. Actions that have a high impact, like repairing an item which saves a value of 32.9kg(based on our sources and calculation), are set to monthly frequency thereby preserving leaderboard fairness and reduces the chance of distortion of cumulative totals. This action maintains system integrity while giving users more motivational incentives.

Challenge table					
Challenge	Description	Carbon Saved(KG)	Frequency	Assumptions	Links
1	Eat a vegetarian meal	0.64103	Daily	Assuming people eat 3 meals a day, rounded to 5.sf	[4]
2	Walk to campus	0.17	Daily	Assuming users would have otherwise drove a petrol car. This figure is carbon emissions saved per Km of walking.	[8]
3	Use a reusable coffee cup	0.026	Daily	Assuming users use cup until end of average lifecycle which is 500 uses	[12]
4	Use a reusable water bottle	0.47123	Daily	Assuming users use bottle until end of average lifecycle which is 500 uses	[7]
5	Recycle	1.9	Daily	Assuming recycled all materials, 61 kg divided by 31 days	[2]
6	Shop for second hand clothes	13.9	Monthly	This figure was generated on general clothing using a linked calculator	[10]
7	Line dry clothes	1.8	Weekly	The average drying cycle releases 1.8 kg's of Carbon emissions	[9]
8	Borrow book from library	0.7	Weekly	N\A	[11]
9	Donate an item to charity	13.9	Monthly	Assuming this is equivalent to buying second hand clothing	N\A
10	Repair an item of clothing	32.9	Monthly	Assuming only one item is repaired	[1]
11	Purchase 2good2go	2.7	Monthly	Assuming each bag on average saves 2.7 kg	[3]

This is a table of the challenges with their respective features like the carbon emissions saved, which taxonomy the challenge falls under and sources of where we found the carbon emission for each task.

Web Design

When planning the design for our front end webpages we used the 4 design principles put forward by the government of the UK to ensure our website is accessible to all users and create a friendly, approachable design.

Perceivable - Images include text alternatives for screen readers, content is logically structured, and the Verdana font is used throughout for dyslexia friendliness. High contrast white text on a dark background ensures readability, and the page supports zooming for low-vision users.

Operable - No flashing content is used to protect users with photosensitive conditions. Navigation tabs are persistently visible with colour-change feedback on interaction, and buttons are sufficiently large and spaced to prevent misclicks.

Understandable - The website is presented in English with a consistent colour scheme and layout throughout. Form fields have clear, visible labels, and login is kept simple, requiring only a username and password.

Robust - The website is written in semantic HTML to ensure compatibility with assistive technologies and across different browsers. All images include alternative text.

In our efforts to create such an accessible interface for our users we came up with the user friendly designs as a Lo-Fi prototype to plan our implementation of the UI. This design is shown in the transition diagram below.

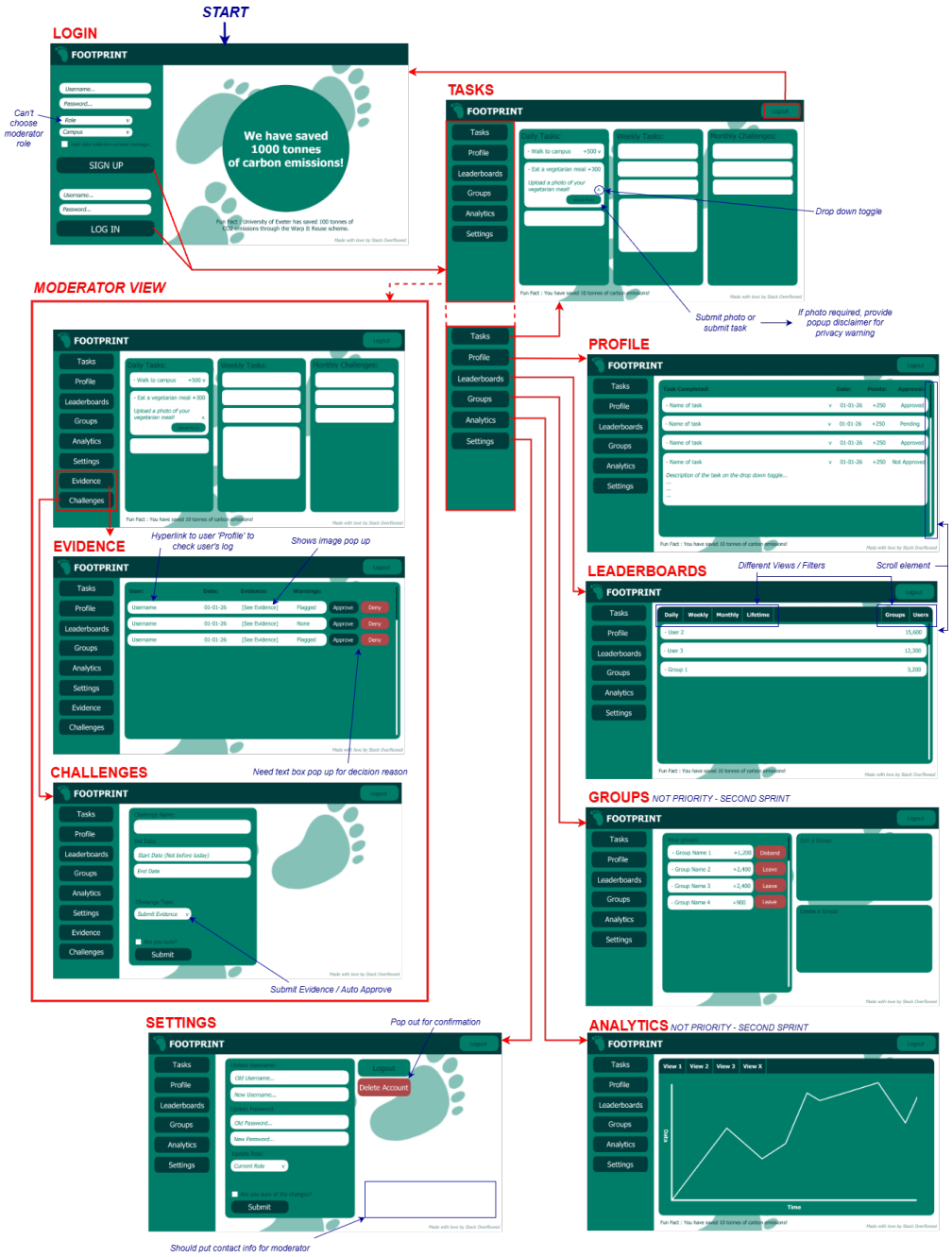

Figure 1: UI Transition Diagram

Architecture and Implementation

The project is built on a range of tools and technologies that collectively enable us to develop, deploy, and interact with the web application. From the user's perspective, the front end serves as the primary interface; allowing users to navigate the application, trigger actions via buttons, and submit data through forms and input fields. Each of these interactions generates a request that is sent to the back end via HTTP fetch and POST methods, where it is processed and the appropriate response is returned to update the interface and deliver the relevant functionality.

These requests are processed and managed by a Java-based back end built around a Controller-Service design pattern, which divides the system into modular layers, each with a distinct responsibility. The controller layer acts as the entry point, receiving requests from the front end and routing them to the appropriate service classes. The service layer houses all core business logic, including evidence handling, moderation, and leaderboard updates, performing the necessary calculations and operations before returning results to the user via the controllers. Underpinning this is the model layer, which defines the structure of data objects as they are stored in the database, and the repository layer, which sits closest to the database and handles all data access operations, allowing the service layer to focus purely on logic rather than database management. This layered, modular architecture makes the system straightforward to test, debug, and extend without disrupting existing functionality

Central to the application is a relational database built around a user-centric design, ensuring that all stored data is associated with a specific user and that each user has an independent experience within the system. The USERS table stores core account information, including usernames, hashed passwords, roles (user or moderator), campus, and year of study. The CHALLENGES table defines each challenge, storing its title, description, point value and other classification data, while the USERS-CHALLENGES junction table links the two, tracking which challenges each user has and has not completed. The EVIDENCE table manages image files submitted for challenges that require proof of completion. The LEADERBOARD is constructed dynamically on request by querying the USERS table, ordering users by points, and returning the top ten results. Finally, a GROUPS table is planned for the second sprint, which will allow users to connect with one another under shared group titles. These tables and their relationships can be seen in the table below.

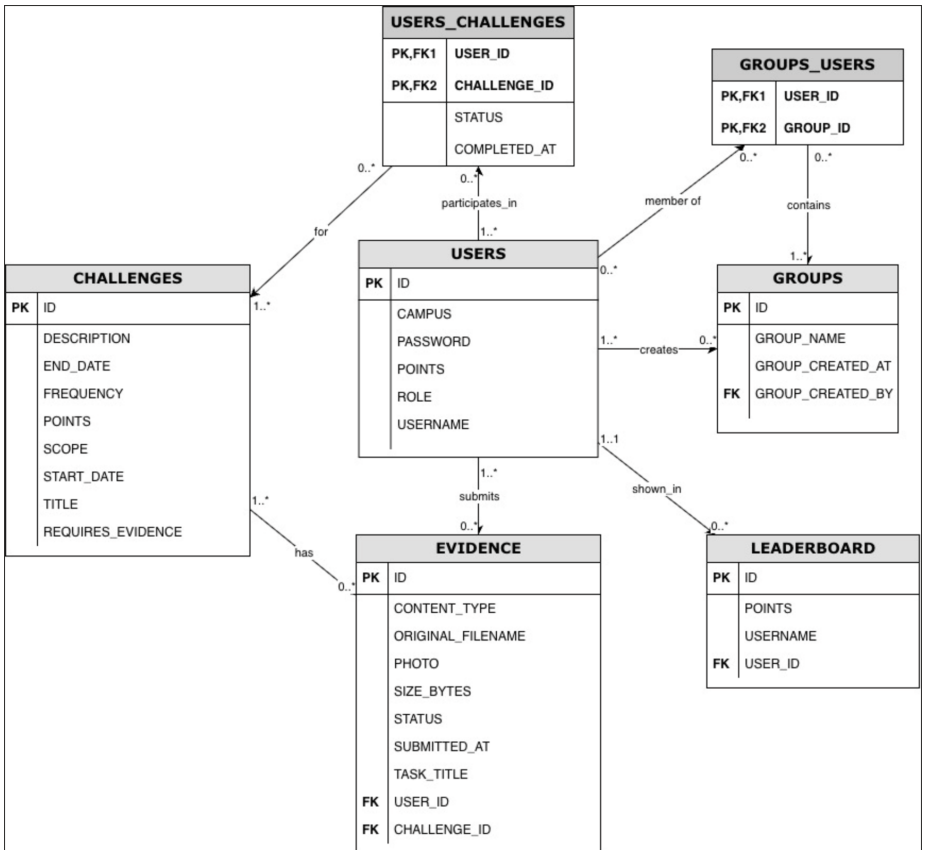

Figure 2: ER diagram

Overall, the architecture underpinning this application has been designed with clarity, scalability, and maintainability in mind. The Controller-Service pattern enforces a clean separation of concerns across the system, ensuring that request handling, business logic, and data access remain distinct and independently manageable. This makes the codebase straightforward to extend, test, and debug as the project evolves. Paired with a user-centric relational database that ties all data to individual accounts, a responsive front end, and a robust Java back end, the system delivers a seamless and engaging experience for users.

Evaluation

When thinking about the evaluation of the implementation, there are many techniques that need to be used to effectively test the entire program. We have opted to use a combination of unit testing, integration testing, and end-to-end (E2E) testing.

JUnit 5 is being used for both unit testing and integration testing as it is very well known and provides easy integration with the Spring framework used to develop the website. Unit testing is used to test the individual methods and their functionality regardless of the database and frontend. Mostly, this is done using the Mockito Testing Framework, which allows us to mock the repository/database classes and control any access to them.

With integration testing, the annotation *@DataJpaTest* on the test classes allows us to auto-wire the repositories, as opposed to mocking them, helping us to test the code and functionality with a functioning repository system. Regarding E2E testing, we selected a vertical slice of all functionalities implemented during the first sprint. This vertical slice reflects the most important features, and they will be manually tested by going through the production website recording successes and failures.

The overall results of the tests helped us to understand what needs to be completed for the next sprint. All unit tests have been passed, giving us good coverage on the classes that were implemented, which is reflected in the final production of the website, where all features that were implemented, work as intended and produce no errors during runtime. The end-to-end test also proves really useful to the evaluation of the implementation, as it clearly shows us what features have passed, partially passed or failed given our criteria for the MVP.

Sprint plan for CW2

In the second sprint, we will implement the analytics tab, which includes graphs and statistics, regarding users' recent activity amongst others. Furthermore, we will be implementing groups where users can create or join a group where some of them will require a specific code. This is done to ensure that societies/departments in the university can only have their own members joining the group. In addition, there will be incentives like streaks and badges to motivate users. The latter will also be able to filter the leaderboard by date. Moreover, additional security measures will be added.

Sprint 2 goals		
1	As a user, I want to easily be able to join/create groups with a code for only members of a specific society to be able to join.	A group system where users can join by having a unique code for each.
2	As a user, I want to be able to see what badges I can receive from challenges and see the badges of other users.	A list of badges users can get for matching a certain criteria and the badge will be present on user profile so they can see and so that other users can see.
3	As a user, in the case my submission is denied, I would like to know why the moderator rejected my submission	In the case moderator deny a request, moderators will return a text explaining why they made the decision to reject it.

Management

To structure the development of our project, we adopted an agile approach with the goal of producing a minimum viable product within our first sprint. Agile was well suited to our timescale, as its short development cycles allow deployable code to be produced quickly and enable workstreams to run in parallel rather than sequentially. We also held weekly meetings to review progress using scrumboards, which helped the team stay coordinated and on track. Together, these practices allowed us to avoid the rigidity of hard methodologies such as the waterfall methodology, where each phase of development must be completed before the next can begin, which would not have been compatible with the time constraints we were working within.

In order to facilitate a better structure for managing responsibilities we designated roles to each member. While we endeavoured to support one another when needed, this structure enabled us to focus on our respective tasks and provide a clear outline on who would be responsible and consulted on different areas of the project, giving us a clearer method to delegate tasks and navigate a roadmap to complete our first sprint.

The following entails the responsibilities and contributions of each member of the team:

Madi — Project Lead Responsible for organising and leading meetings, ensuring final product quality, helping facilitate the requirements and design of the project and liaising with all the members and facilitating communication between the team. Acted as scrum master, managing the scrum boards and maintaining the risk register to avoid a variety of issues from arising such as scope creep. Also provided support to UX design by creating the digital designs for the UI assets and layouts.

Ben — Technical Lead In charge of managing the architecture and quality of the code, ensuring that it is maintained for future developer use. Delegated tasks to members of the development team based on priority and individual strengths, while also taking on development tasks where needed on all elements of the code.

Jai — Data Lead Managed the structure and design of the database, and how the data will be securely stored. Supported documentation in conjunction with this.

Marko — Developer Completed development tasks on all elements of the code to incorporate different features and functionalities to the system as well as establishing security systems across the project.

Phoebe — Developer Provided important analytical information from research to facilitate the data used in challenges and tasks, and worked on the HTML as well as CSS elements of the code.

Zarreen — UX Design Considered all the legal and ethical aspects of our product, including our license justification. Ensured that the standards we have set are upheld across our code by coordinating with the development team. Worked on design aspects of the project, keeping accessibility and stakeholders in mind, and helped produce the demo slides for our team presentation.

Davi — Testing Facilitated all of the testing suites and implementations, coordinated with developers to make sure testing coverage was broad and that the final product was bug free and effective. Also provided support in documentation and code.

Armin — Documentation Responsible for the report of our first sprint prototype, has coordinated with other members of the group to ensure information is accurately documented and helped with database design.

For a full list of logs and contributions made by each member please see the second part of our `meeting_minutes.pdf` in the directory `2_process`.

Risk Management

When producing a large-scale product for a client, it's easy to overlook vital risks when focusing solely on the criteria required to produce the final outcome. Once challenges arise and appropriate mitigation or contingency isn't put in place it can require more resources to be drained to amend these issues. Common points of failure for a project can come from a lack of testing (R7, R16) when not considered in advance of development leading to problematic bugs the client encounters. Other areas of failure may originate from a lack of clear definition in requirements when understanding the problem (R1, R8, R10) or they may stem from within the functionality of the team itself (R3, R9, R18).
In order to minimise risks, a risk register was maintained through the first sprint to cover all these individual risks that were owned and mitigated by respective members of the team. This provided a useful ongoing log to track potential issues that would arise and pre-emptively respond to minimise potential negative outcomes by putting mitigation in place to aptly prevent it occurring.
For the risk register please find it in the directory `2_process`.

First Sprint Strategy Review

After reviewing our first sprint we believe that we collaborated effectively and efficiently together in order to create a strong minimal viable product for our first sprint. This now means that we can focus on our goal of engagement in our second sprint with this foundation now laid in place, while we also conjunctively make improvements to our current implementation along the way.
Discussing with the team on our collaborative efforts, we want to move forward in the same process we did for sprint one, focusing on our shorter weekly iterative sprints and delegating our tasks weekly. However we would like to revise our risks more thoroughly. Certain risks such as R20 and R21 only became prevalent when reviewing our sprint based on unforeseen circumstances and perhaps could have been mitigated sooner had the risks been revised and discussed more frequently in the team. Given the time scale we have to achieve this project, our team wants to prioritise managing our scheduling and deadlines effectively to create the best product, which at its core means handling our risks as adeptly as possible.

Sprint 2 Introduction

This section outlines the key developments completed during Sprint 2. Building on the minimum viable product established in Sprint 1, we significantly extended Footprint's functionality across four core areas: user engagement, personalisation, security, and platform ethics. The primary objectives for this sprint were to introduce groups, badges, and streaks, develop an analytics dashboard, refine the user profile experience, and implement additional security and ethical safeguards. All planned features were successfully built and tested within the sprint window. As in Sprint 1, the team followed an Agile Scrum methodology; maintaining weekly meetings, keeping the Scrum board up to date, and distributing responsibilities clearly across team members.

1 Requirements

This is a list of the requirements implemented during sprint 2. Using MoSCoW, which was used in Sprint 1, we were able to have our user stories with their acceptance criteria.

Sprint 2 Requirements		
#	User Story	Acceptance Criteria
1	As a user, I want to easily be able join/create groups/societies with a code for only members of a specific society to be able to join.	Users can access a tab where users can join a group with a specific group code, and users can create a group where they will be given a code so other users can join by entering that code in the join group section.
2	As a user, I want to be able to see a carbon emissions reduction analytics dashboard so that I can track my environmental impact over time.	A section where users can view a pie chart and a graph of their activities and how much Carbon was saved over a period of time(Week, Month, and all time) and pie chart showing users which activity has saved the most carbon emissions.
3	As a user, I want to know what badges I can receive and easily see the badges of others	A section where all the badges users can receive are on display and if user clicks on a badge, a pop up is returned with the criteria the user has to achieve in order to get the badge, once a user has been awarded a badge, they can use it on their profile and will appear next to their username when they are on the leaderboard which other users can see.

Sprint 2 Requirements		
#	User Story	Acceptance Criteria
4	As a user, I want to see my streak	StreakService calculates completion, badge threshold for this at 7/50/100 days
5	As a developer, I want to increase security so users can trust the platform	Passwords require a minimum of 8 chars, SQL injection protections are in place, username and group names must be unique with no special characters. Evidence images are manually deleted after 1 year.
6	As a developer, I want to ensure that the website is ethical and transparent so users are clearly informed about important policies.	Users have links to legal documents and terms and conditions and an accessibility statement. Moderators must provide a reason for denying a user submission
7	As a user, I want a profile tab so i can see all my actions and evidence approval status	A profile tab where users can see their completed tasks and submission statuses, includes how many points were appointed for each task and if a task was denied there is a section to explain the reasoning.

2 Architecture and Implementation

Sprint 2 extends the existing Controller-Service architecture established in Sprint 1 with new modules. The design pattern has been preserved, with new controllers, services, models and repository instances introduced for each feature.

2.1 Badges

A badge system has been implemented to reward users upon reaching specific achievement milestones, built around two database tables. The BADGES table records all badges a user has earned, with each row representing a single award, while the USER_BADGES table tracks each user's currently selected badge for public display. Eligibility criteria are defined within the BadgeService class as hardcoded thresholds, covering metrics such as points earned, group memberships, and streak length. When a user satisfies the criteria for a given badge, BadgeService.awardIfEligible() automatically grants and records it in the BADGES table, with database constraints in place to prevent duplicate awards.

Once a badge has been earned, users may select it from their collection to display alongside their username on the leaderboard, with their choice persisted in the USER_BADGES table. BadgeService.evaluateAllBadgeMechanisms() is invoked following every successful challenge completion, evidence approval, and travel log submission, assessing all 19 available badges against the user's current progress. Most categories follow a three-tier progression structure, such as Trailwalker, Hiker, and Pathfinder for streaks, or New Leaf, Growing Leaf, and Thriving Leaf for daily tasks. Users who earn every other badge are awarded the Footprint Master badge, which serves as the system's highest distinction.

Badge's table			
#	Category	Badge Name	Trigger Condition
1	Points	eco_cadet	Total points ≥100
2	Points	eco_soldier	Total points ≥500
3	Points	eco_captain	Total points ≥1000
4	groups	social_bee	Member of ≥3 groups
5	groups	queen_social_bee	Member of ≥ 10 groups
6	Streak	trailwalker	Daily completion streak ≥ 7 days
7	Streak	hiker	Daily completion streak ≥ 50 days
8	Streak	hiker	Daily completion streak ≥ 100 days
9	Daily tasks	new_leaf	10 ≥ accepted Daily challenge completions
10	Daily tasks	growing_leaf	50 ≥ accepted Daily challenge completions
11	Daily tasks	thriving_leaf	100 ≥ accepted Daily challenge completions
12	Weekly tasks	tree_tender	≥ 5 accepted Weekly challenge completions
13	Weekly tasks	tree_planter	≥ 25 accepted Weekly challenge completions
14	Weekly tasks	tree_gardener	≥ 50 accepted Weekly challenge completions
15	Evidence	amateur_ecologist	≥ 5 approved evidence submissions
16	Evidence	field_ecologist	≥ 10 approved evidence submissions
17	Evidence	certified_ecologist	≥ 25 approved evidence submissions
18	Taxonomy	versatile_achiever	Completed a challenge in every taxonomy category
19	Mastery	footprint_master	All 18 other badges are earned

2.2 Groups

The groups feature allows users to create or join societies and departments within the platform. Upon creation, each group is assigned a unique 8-character alphanumeric invite code generated from a UUID, ensuring access remains restricted to invited individuals only. The GROUPS table stores group metadata including the invite code, group name, and owner reference, while the GROUP_MEMBERS join table manages membership through a ManyToMany relationship. The GroupController handles all group operations: creating new groups, which automatically adds the creator as a member, joining groups via the invite code, leaving groups for regular members, and disbanding groups, which is restricted to the group owner.

When a user joins a group through GroupController.addUserToGroup(), BadgeService.evaluateAllBadgeMechanisms() is automatically triggered to assess group-related badge eligibility, such as the Social Bee badge which requires membership in three or more groups. All group data access is managed through the GroupRepository interface using Spring Data JPA.

2.3 Streaks

The streak system tracks consecutive days of user activity based on accepted evidence submissions. Rather than persisting a streak value directly in the database, StreakService.calculateStreak() dynamically computes the current streak by examining the submission dates of all accepted evidence for a given user. The streak is determined by counting consecutive calendar days working backward from today, where each day must contain at least one accepted submission. If no accepted evidence exists for the current day, the calculation begins from the previous day, and the streak resets to zero as soon as a gap is encountered.

The StreakController uses a @ModelAttribute to expose the streak value across all views, calculating it on demand for authenticated users. The resulting streak count is displayed on the user's tasks page and feeds directly into the badge system, where it is used to evaluate streak-based badge eligibility.

2.4 User Profile

A profile page has been introduced to consolidate all user-specific information in a single location. The page presents the user's complete evidence submission history in a table, displaying the task name, submission timestamp, points awarded, approval status (Pending, Approved, or Rejected), and any rejection reason provided by a moderator. Submissions are listed in chronological order, allowing users to easily track their completed tasks, outcomes, and any feedback received.

The profile also displays the user's selected badge alongside their username in the top navigation bar, provided a badge has been chosen. The page is served through the ProfileController, which queries all evidence entries associated with the authenticated user and passes them to the view for rendering.

2.5 Analytics Dashboard

The analytics tab provides users with data visualisations of their environmental impact and activity over time. The page presents two interactive charts, each filterable by timeframe and scope. A pie chart displays a category breakdown by challenge taxonomy, highlighting which category the user contributes the most carbon savings to. A line chart displays carbon emissions saved over time, with three timeframe options available: weekly, monthly and all time.

Users can also select the scope of their analysis from three levels: individual, year group, and campus, showing the cumulative data of all users in that scope. The AnalyticsController aggregates accepted evidence submissions based on the selected filters, with the EvidenceRepository supplying the underlying data including user ID, taxonomy, and submission timestamp. All charts are rendered using Chart.js and update dynamically as users adjust the timeframe and scope dropdowns.

2.6 Sprint 2 Review

A number of security and ethical improvements were delivered in this sprint. From password validations which now enforces users to have a password of minimum 8 chars, SQL injection protections have been strengthened across all input handling endpoints through the use of parametrised queries and input sanitation. Usernames and Group names are now validated for uniqueness at creation time. Evidence images stored in EVIDENCE tables are subject to be manually deleted in one year from submission.

For the ethical side of things, users that are on the login page are presented with links to terms and conditions and accessibility statement, which both must be consented to before registration is completed. Acceptable use statements are also added to guide users on appropriate content for uploads for evidence and the naming of usernames and accounts. Moderators are now required to provide a written reason for denying a user submission, this ensures transparency and fairness.

2.7 Evaluation

The same testing strategy used in sprint 1 was also used in sprint 2, combining unit testing, integration testing and end to end testing to ensure a wide converge of all the newly implemented features.

Unit testing were written for all new service classes which includes GroupSerivce and the updated UserService streak logic. As in sprint 1, the Mockito framework was used to isolate service behaviour from the database layer, and all the unit tests passed successfully. Integration testing using @DataJpaTest were used to validate the new repository interfaces, including BADGES, USER_BADGES, AND GROUPS_USERS tables. this was used to confirm correct data resistance and retrieval behaviour.

End-to-end testing was again performed using a vertical slice of most important user journeys that were introduced in this sprint. This included creating and joining a group with the group specific code, earning a badge and then displaying this badge, being able to track streaks, navigating the analytics tab with the date and scope (year group, individually and campus) and verifying that moderators are prompted for a reason if they deny a request for a submission. All of these cases have passed with the website and the website functioning as intended along side not producing any any errors during the runtime.

Security enhancements were validated through targeted testing, SQL injection attempts against key input fields were blocked. The password minimum length constraint was confirmed to be enforced on the client side and server side. Uniqueness validation for usernames and group names was verified by attempting to register duplicated values, this passed as they if a user enters a username or groupname that is already in use they are returned with an error message.

3 Final Evaluation

3.1 Sprint 2 management

Our management strategy for sprint two remained mostly unchanged, the biggest adjustment was migrating our scrum board to solely be on Github so that we could have more proactive collaboration, as well as making use of the tool's functionality. This also allowed members of the team to update tasks at their own discretion.

Discussing our project management strategy at the sprint one review allowed our Project Lead to focus on continuing methods that worked well such as weekly meetings and weekly delegation of tasks, as well as providing adjustments where other parts of the strategy had failed to be effective such as fostering more collaborative and open discussions to prevent miscommunication, and ensure the clarity of task objectives.

3.2 Risk Management

In order to keep on top of risks, we reviewed our risk register at the end of the first sprint, and collaborated more openly as a team in our second sprint to navigate any mitigation solutions. This enabled us to deal with risks more effectively, such as R21, where some team members had felt they had been undertaking higher workloads which they were unable to manage. Reviewing this risk as a team, allowed us to discuss sharing workloads and how each team member could be supported if necessary and who would most likely need that support.

We were also able to reflect on risks that hadn't been adequately mitigated, and instead of finding a one-time solution, creating ongoing solutions that were upheld through the second sprint. One such example is R6, where our UX Lead was challenged with a lack of communication from the developers in order to verify ethical standards outlined were met, so to improve this new ongoing mitigations were added to create a two-way communication as weekly meetings, allowing both parties to work more effectively.

3.3 Success measures and methods

To provide our product with measurable success, we produced an evaluation form which was distributed to potential consumers. The form allowed participants to answer questions regarding their interactions with various elements on the website from both security and accessibility standpoints. Additionally, engagement questions were included to produce an understanding of where users were most interested in the website. Finally, users were asked about how many bugs they had encountered as well as allowing them to leave worded feedback on bugs encountered or how they would improve the website.

Reviewing the responses from the form, we were able to assess that the majority of users engaged on the platform with a mobile device (75%), a large

proportion of which reported issues with navigating the mobile layout, so modifications were made to ensure the readability and intuitivity of the layout. Badges and analytics were the most favourable features of the website (28.6% and 21.4% respectively), which highlights the importance of the gamification elements as well as the visualisation of the carbon savings. Carbon accuracy was assessed by cross-checking challenge values against their sources and verifying normalisation to CO_2.

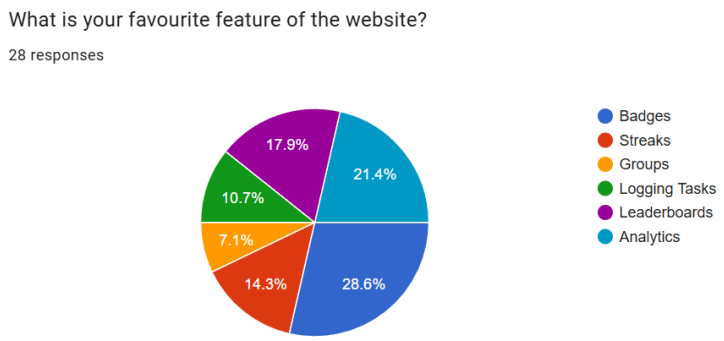

Figure 3: Survey results for users favourite features

75% of users reported encountering no bugs in the system, with the following 14.3% reporting very minimal bugs encountered. Achieving minimal bugs in our system allows us to reflect successfully on our goal of maintaining runtime stability. As we received worded feedback, we were able to the bugs that were reported within the responses and fix any issues that were overlooked in our testing, one such example of this was found in the travel feature. Users were able to enter arbitrarily large values, which caused the stored points figure to exceed the maximum size of a signed 32-bit integer, producing a large negative number which shouldn't be possible. This was fixed by introducing boundaries and limits to the functionality through moderating evidence submitted for journey over a certain distance and limiting the number of times users can log a journey.

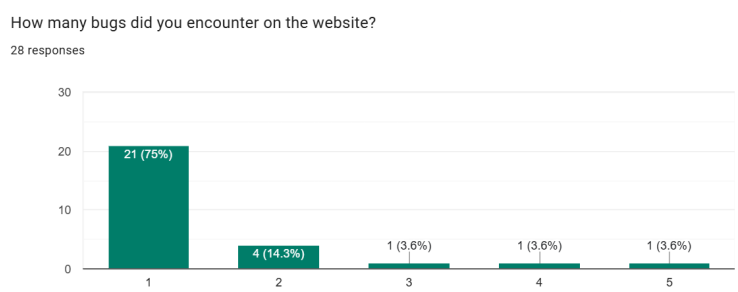

Figure 4: Survey results for number of bugs found by users

3.4 System Limitations

The Footprint application is hosted on Render's free tier, but while this arrangement is sufficient for demonstration and coursework evaluation purposes, it introduces several constraints that are worth acknowledging. The most consequential of these is Render's spin-down behaviour for free-tier web services. When the application receives no traffic for a sustained period, the hosting container is automatically shut down. As a result, the first visitor to arrive after a period of inactivity will encounter a cold start delay of up to 60 seconds before the login page becomes accessible. Beyond this, the free tier imposes strict resource limits on CPU and RAM allocation, a relevant concern given that the platform incorporates multi-user features such as leaderboards, analytics, and group management.

The PostgreSQL database instance is likewise provisioned on Render's free plan, which enforces an approximate storage ceiling of 1 GB and may be subject to automatic expiry following 90 consecutive days of inactivity. This introduces a degree of risk to long-term data persistence that would be unacceptable in a production environment. Furthermore, the free tier provides no custom domain support, no service-level agreement guarantees, and no facility for persistent background processing, meaning any features that would benefit from scheduled jobs or asynchronous tasks are effectively constrained by the platform. Taken together, these limitations make the current deployment unsuitable for production-scale or long-term real-world use without migration to a paid hosting tier.

Another notable limitation of our evaluation is the absence of a prolonged deployment period. While we were able to gather genuine user feedback and measure engagement, we were unable to assess user retention, which would offer considerably deeper insight into the long-term effectiveness of the platform. Given additional time, measuring retention would be our next evaluative step, with the findings used to inform future feature development.

A further limitation concerns the carbon model, which relies on a number of simplifying assumptions. These range from fixed meal counts and average vehicle emissions to standardised product lifecycles, each of which reduces individual accuracy. As a consequence, the platform is currently scoped for use within a limited geographic area, and the underlying challenge data would require recalibration before the application could be deployed more broadly.

3.5 Limitations of the Incentive Structure

The application's incentive structure introduces several unintended consequences that warrant consideration. The evidence submission system is vulnerable to abuse, as users may resubmit photographs from previous submissions to fraudulently claim points. While moderator review provides a degree of mitigation, it cannot be considered a fully reliable safeguard against this behaviour. Users optimising purely for leaderboard rank may also preferentially select frequent, low-impact challenges over more environmentally significant monthly tasks, though per-challenge submission limits partially address this tendency.

Beyond gaming the system, the competitive leaderboard risks disadvantaging users subject to financial or physical constraints. A user with limited disposable income may be unable to participate in certain challenge types, while a physically disabled user may be required to use motorised transport to campus, precluding participation in active travel challenges entirely. Future development should therefore prioritise ensuring that the reward structure remains equitable and inclusive for all users, regardless of their individual circumstances.

3.6 Environmental impact

If users are able to adopt the encouraged behaviours, the carbon emissions saved could be significant, for example if 100 users walked to campus, and assuming the average distance students walk to campus and back is 5km (this figure is from [6], the average walking distance for students in Valencia and assuming they are walking back too, this figure can be different for student at the University of Exeter), the total carbon emissions saved would be 2500 CO_2 for the day. Although the platform itself would have its own digital footprint, the infrastructure used to run this offers limited transparency on energy sourcing so we will not be able to account for the potential carbon emissions released from this.

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